CareCompass

Team Diligence | IIT Patna | Version 0.2 | 24 October 2026

Broad complaint intake, controlled care navigation

CareCompass accepts text or voice in English, Hindi and Hinglish, extracts the presenting complaint and asks one decision-relevant follow-up at a time. Version 0.2 expands beyond fever to 17 named complaint families plus an other fallback. Stomach pain now receives relevant severity, warning-sign, duration and location questions rather than a generic service menu. The system does not diagnose, prescribe, make real bookings or dispatch caregivers.

Shared API-first implementation

FastAPI and Pydantic enforce the organiser response schema. The scoring API, UI adapter and predictions exporter use the same core: replay user evidence, run local safety checks, optionally extract facts with Groq Account A, recheck safety, choose an approved question or route, then filter and rank caregivers. UI-only evidence and concern metadata remain outside the scoring response. A native mobile-first HTML/CSS/JavaScript interface avoids a frontend build layer; Docker includes FFmpeg.

Expanded extraction and adaptive questions

The shared local/live extraction schema includes complaint category and text, severity, pain location, sudden onset, duration including hours, progression, patient age, temperature/currentness, associated symptoms and evidence-backed clinical findings. Unknown, denied, historical and hypothetical evidence remain distinct. Short answers are interpreted using the preceding question, including reverse polarity such as whether fluids can be kept down. Already supplied facts are skipped; normally no more than four follow-ups are asked.

Associated focal complaints receive immediate safety screening without transferring their duration or severity into the main complaint. Sequential patient changes reset state. Complex simultaneous complaints and ambiguous patient histories remain limitations requiring human assessment. Unfamiliar complaints use focused general clarification and an honest fallback; broad text acceptance is not universal medical competence.

Two separate server-side Groq accounts

Account A uses configurable openai/gpt-oss-20b strict structured extraction. Account B uses configurable whisper-large-v3 transcription; a benchmark script compares it with whisper-large-v3-turbo. Explicit independent clients bind different keys. Credentials never enter frontend code. Exact evidence quotes and basic numeric grounding are checked. Bounded retries, deadlines and malformed-output handling prevent silent provider failure. The backend loads .env automatically without executing it as shell code.

Without keys, text uses a labelled local rules baseline and voice reports unavailability. Account separation and provider responses are transport-stub tested; live Groq calls and comparative transcription quality have not been measured here.

Safety and taxonomy gaps

An affirmed active controlled warning sign forces emergency_112/emergency, null follow-up and no matches. Local checks run before network calls, after extraction and after every answer. Version 0.2 adds selected abdominal, headache, airway, vision, confusion, urinary and back-pain warning-sign pathways. Where no controlled flag accurately fits, actual evidence is preserved in reasoning and the emergency response has an empty red_flags list. This documented contract-gap assumption requires organiser confirmation.

Prompt in-person assessment concerns are surfaced immediately, even while a safety-changing question is asked. Unresolved critical information or urgent assessment needs suppress routine booking. The supported same-day clinician-consultation category is a coordination fallback, not a claim that virtual care replaces necessary in-person assessment. No disease probability, platelet inference, prescription or dispatch claim is made.

Deterministic matching

Eligibility requires active status, a compatible role, explicit procedural skills, service radius for home visits and remaining availability at an injectable IST reference clock. Haversine distance is ignored for virtual doctors. Ranking uses README weights: language 0.35, proximity 0.30, skills 0.20 and rating 0.15, with stable ties. Future availability is unconfirmed and booking disabled. Insulin is not matched to IM staff without explicit eligibility evidence.

Measured results and trade-offs

The supplied 340 examples contain only 66 literal messages, empty histories and repeated templates. Thirty-eight emergency rows use unsupported flags; proposed corrections remain separate. All-data rules results: care macro-F1 0.8219, urgency macro-F1 0.8514, emergency recall 56/56 and schema compliance 1.0. Matching label-agreement proxies on 209 nonempty-reference rows: Hit@1 0.4689, Recall@3 0.5478 and NDCG@3 0.5077. A published 103-row group-disjoint subset and complete confusion matrices are included.

The earlier narrow version scored 0.8698/0.8736 care/urgency F1. Version 0.2 asks relevant safety questions on some cases that labels route immediately, reducing label agreement while broadening behaviour. Rules were authored after dataset inspection: neither evaluation is blind hidden-test evidence or clinical validation. No official aggregate score improvement is claimed.

114 Python tests passed, including independently phrased complaint, multilingual, negation, late-emergency, correction, outage, eligibility and audio-validation cases. A real-HTTP DOM integration check passed the stomach flow, choices, matching, simulated booking, Hindi and corrected-transcript safety. DOM tests do not establish browser visuals or microphone permission behaviour. Live Groq, real recordings, Docker execution and clinical mentor review remain outstanding.

Sources, attribution and disclosure

Sanocare synthetic care-triage dataset v1 (2026), supplied for IIIT-Delhi; FastAPI, Pydantic, Uvicorn, HTTPX, Starlette, FFmpeg, python-dotenv, pytest and optional jsdom; Groq-hosted GPT-OSS and Whisper. OpenAI ChatGPT assisted code, tests, analysis and documentation. Primary guidance: nhs.uk/symptoms/stomach-ache; nhs.uk/symptoms/headaches; nhs.uk/conditions/back-pain; nhs.uk/conditions/anaphylaxis; nhs.uk/symptoms/eye-pain; www2.hse.ie/conditions/stomach-ache. Full source-to-rule mappings and limitations are in docs/BROAD_INTAKE.md. Synthetic prototype only.